
GROUP INFORMATION

Group: _____

Coach: _____

Date: _____

Course: Data Science

OBJECTIVE

You will analyze a video game sales dataset using Excel / Python.
You will perform exploratory data analysis (EDA) and based on it make a recommendation for indie game studio.

DATASET INFORMATION

Dataset: Video Game Sales

Variables:

- Name
- Platform
- Year_of_Release
- Genre
- Publisher
- NA_Sales
- EU_Sales
- JP_Sales
- Other_Sales
- Global_Sales

STEP 1: LOAD & INSPECT DATA

Goal: Understand dataset structure

Tasks:
[ ] Open dataset and convert to table (Excel) / make dataframe (Pandas)
[ ] Inspect column names and data types
[ ] Identify missing values
[ ] Perform required data cleaning and preparation

Notes (data issues found):

STEP 2: DESCRIPTIVE STATISTICS & OUTLIERS

STEP 2.1 SUMMARY STATISTICS

Goal: Understand central tendency

Fill in values:

Mean: _____

Median: _____

Standard deviation: _____

Min: _____

Max: _____

Q1: _____

Q3: _____

STEP 2.2 DISTRIBUTION ANALYSIS (RAW VS LOG)

Tasks:
[ ] Create histogram for Global_Sales
[ ] Create new column: Log_Sales = LOG(Global_Sales + 1)
[ ] Create histogram for Log_Sales

Questions:

1. How does the distribution change after log transformation?

2. Are extreme values still dominant after transformation?

3. Which representation is more balanced and why?

TOP 10 GAMES BY GLOBAL SALES:

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____

Do these games still dominate after log transformation? Explain:

DECISION QUESTION:

Which should be used?

[ ] Raw data
[ ] Log data
[ ] Both

Justification:

STEP 2.3 OUTLIER DETECTION

IQR: _____

Lower bound: _____

Upper bound: _____

Number of outliers: _____

STEP 2.4 OUTLIER INTERPRETATION

Are outliers meaningful or errors?

STEP 2.5 FINAL DECISION

How should outliers be handled?

[ ] Keep

[ ] Remove

[ ] Use median

[ ] Other

Justification:

STEP 3: TIME TREND ANALYSIS

Tasks:
[ ] Pivot table (Year_of_Release vs Global_Sales)
[ ] Line chart

Peak years:

STEP 4: PLATFORM ANALYSIS

Top platforms:

1.
2.
3.

STEP 5: GENRE ANALYSIS

Top genres:

1.
2.
3.

STEP 6: PLATFORM × GENRE ANALYSIS

Key observation:

STEP 7: SIMPSON'S PARADOX

What contradiction did you observe?

STEP 8: VISUALIZATION

Which chart was most useful?

[ ] Stacked bar chart

[ ] Grouped bar chart

[ ] Other

Why?

STEP 9: BUSINESS INSIGHTS

1. Platform insights:

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2. Genre insights:

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3. Simpson's paradox insight:

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4. Marketing recommendation:

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5. Risk of aggregation:

STEP 10: INDIE STUDIO STRATEGY

Primary niche genre:

Secondary genre:

Justification:
